

An Immunological Approach to Dynamic Behavior Control for Autonomous Mobile Robots

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Abstract

Conventional AI (Artificial Intelligence) technology has been criticized on many grounds such as brittleness under dynamically changing environments. To overcome this problem, another approaches called New AI, Emergent Computation, Animat Approach, Behavior-Based AI and so on were proposed and confirmed their usefulness. On the other hand, biological information processing systems have many interesting functions and are expected to provide various feasible ideas to engineering fields, especially robotics. Immune system plays important roles to maintain its own system against dynamically changing environments. Therefore, we believe that immune system would provide a new paradigm suitable for dynamic problem dealing with unknown environments rather than static problem. In this paper, we propose a new approach to behavior-based AI by paying attention to biological immune system. We confirm the feasibility of our proposed method by applying to a behavior control of an autonomous mobile robot in cluttered environment as a practical example.

1. Introduction

Conventional artificial intelligent systems based on a functional decomposition, leading to a so-called "sense-model-plan-action" cycle have been criticized on many grounds over the last decade. Typical criticisms are that these systems show brittleness for environmental changes, and required much computing time for mapping complex sensory inputs into complex internal models before action can be taken. Therefore, in recent years much attention has been focused on the *reactive planning systems* (e.g., behavior-based AI, new AI, emergent computation, animat approach, and so on), which have already been demonstrated robustness and flexibility against dynamically changing world[1]-[3].

On the other hand, biological information processing systems such as human beings have many interesting functions and are expected to provide various feasible ideas to engineering fields, especially robotics. Biological information system in living organisms can be mainly classified into the following four systems: (1) *brain-nervous system*, (2) *genetic system*, (3) *endocrine system*, and (4) *immune system* (see Fig.1). Among these systems, brain-nervous and genetic systems have already been applied to engineering fields by modeling as neural

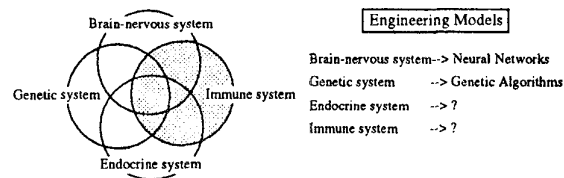


Fig.1: Biological information processing systems.

networks[4] and genetic algorithms[5], and they have been widely used in various fields. Little attention, however, has been paid to application of the other systems (i.e., endocrine and immune systems) notwithstanding their important characteristics.

Immune system, in particular, have various interesting features such as immunological memory, immunological tolerance, micro-pattern recognition, non-hierarchical distributed structure, and so on viewed from the engineering standpoint. In addition to this, recent studies on immunology have clarified that the immune system does not only detect and eliminate the non-self materials called *antigen* such as virus, cancer cells and so on which come from inside and outside of the living system, but plays important roles to maintain its own system against dynamically changing environments. Therefore, immune system would be expected to provide a new paradigm suitable for dynamic problem dealing with unknown environments rather than static problem. So far, several methods have been proposed to realize phenomena of the immune system from the standpoint of mathematical analyses[6][7]. However, the immune system has little been applied to engineering fields in spite of its productive characteristics.

Based on the above facts, we have been trying to engineer methods of the immune system, and apply to robotics and so on[8][9]. We expected that there would be an interesting AI technique for dynamically changing environment by imitating the immune system in living organisms.

In this paper, we develop a new behavior-based AI technique that can cope with a dynamic and complicated environment from the immunological standpoint inspired by *Maes's behavior networks*[10][11]. The advantage of our method is to be able to comprehend a behavior-based AI technique uniformly using immune dynamics. To confirm the feasibility of our proposed method, we take the behavior control of autonomous mobile robots as a

practical example and compare our proposed method with Maes's behavior network approach applying to this example.

2. Immune Networks

2.1 Overview of Immune System

The basic components of the immune system are *lymphocytes* that are classified into two types, namely *B-lymphocytes* and *T-lymphocytes*. B-lymphocytes are the cells produced by *bone marrows*. Roughly 10^7 distinct types of B-lymphocytes are contained in a human body, each of which has distinct chemical structure and produces many "Y" shaped *antibodies* from its surfaces. The antibody specifically recognizes an *antigen* like a *key and lock relationship*. The structure of the antigen and the antibody is shown in Fig.2. In this figure, the relationship between the antigen and the antibody is illustrated. The portion on the antigen recognized by the antibody is called *epitope* (antigen determinant), and the one on the antibody that recognizes the corresponding antigen determinant is called *paratope*. However, regarding immune response just as key and lock relationship between antibodies and antigens is an old-fashioned interpretation of immune system.

Recent researches on immunology have clarified immune system is not so simple. Namely, antibodies also have antigenic characteristic called *idiotope* (see Fig.2) and key and lock relationship also exists between different species of antibodies.

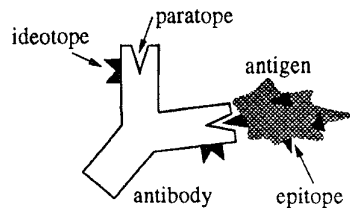


Fig.2: Structure of antigen and antibody.

2.2 Jerne's idiotypic network (immune network) hypothesis

Based on the above fact, *N.K.Jerne* who is an immunologist proposed a remarkable hypothesis: *idiotypic network hypothesis*[12]-[14]. This hypothesis explains that antibodies do not exist independently in living organisms, but communicate with each other through *idiotope* and *paratope*. This idea is schematically shown in Fig.3.

In this figure, the idiotope Id1 of the antibody 1 stimulates the B-cell 2, which attaches the antibody 2 to its surface, through the paratope P2. Viewed from the standpoint of the antibody 2, the idiotope Id1 of the antibody 1 works simultaneously as an antigen. As a result, the B-cell 1 with the antibodies 1 are suppressed by the antibody 2. On the other hand, the antibody 3 stimulates the antibody 1 since the idiotope Id3 of the antibody 3 works as an antigen for the antibody 1. These

mutual interactions (stimulation and suppression) chains between different species of antibodies form the large-scale closed chain loop which works as a self and non-self recognizer. The heart of Jerne's idea is that the immune system does not act as a unit-level recognition system but as a system-level recognition system.

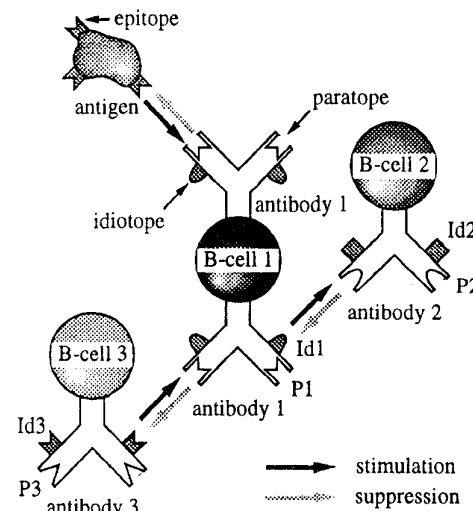


Fig.3: Jerne's idiotypic networks.

3. Proposed Behavior Control System

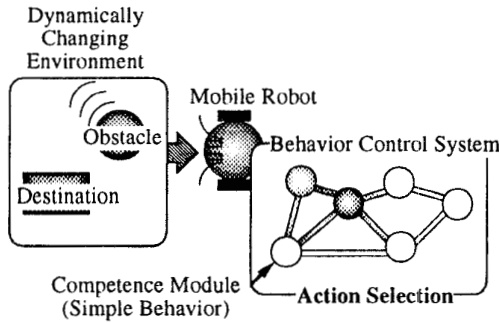
3.1 Action selection

In recent artificial intelligent technology, fatal limit of conventional approaches based on the functional decomposition have been pointed out, therefore, much attention has been focused on the approaches based on behavior which demonstrate robustness and flexibility against dynamically changing environments. However, the arbitration between competence modules in these approaches encountered its difficulty. Among many behavior-based approaches, Maes proposed an interesting method: *behavior network system*, under which the action suitable for the current situation and the given goal emerges as the result of interaction of competence modules, that is, activation flows through many simple autonomous agents[10][11]. However, in her method, difficulties of adjustments of parameters and design of agents have been pointed out. Moreover, there is also a problem how to prepare initial agents to cope with the unknown future events.

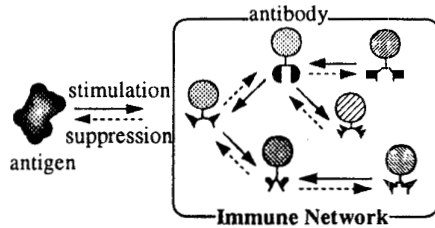
3.2 How do we connect robot behavior control with immunology?

In this study, we attempt to construct a behavior control of autonomous mobile robots from the immunological standpoint, namely using immune networks. Fig.4 shows the correspondence between the behavior control of autonomous mobile robots and the immune network system, schematically. As shown in

this figure, information on surrounding environment of the robot, for example, distance between obstacles and robots and directions of obstacles and destination work as antigens, and competence module (simple behavior) and interaction between behaviors can be regarded as antibody (or B-lymphocyte) and stimulation/suppression between antibodies, respectively. The basic principle of our proposed method is that the immune system equipped with the robot select the competence module (antibody) which is most suitable for the current situation (antigen).



(a) Robot control with action selection mechanism.



(b) Immune network.

Fig.4: Basic concept of immune network-driven autonomous mobile robots.

Furthermore, to make the system balance with current situation and purpose, we divide the immune networks of the autonomous mobile robot shown in Fig.4 into two layers (networks) as in Fig.5. For the sake of convenience, we call these two networks *situation-oriented* and *goal-oriented immune networks*. In the situation-oriented immune network layer, antigen has the information on the current situation such as the existence of obstacles and the direction of destination. On the basis of the structure of the antibody as shown in Fig.2, we define paratope and idiotope as follows: the paratope represents conditions that have to be fulfilled before the antibody can be selected, and the idiotope represents the expected situation of the antibody concerned implements its own action to real world. From these arrangements, the interaction (i.e., stimulation and suppression) between paratope and idiotope is generated based on the chain of causality. On the other hand, in the goal-oriented immune network layer, antigen has the information on given demand such as approaching goal and so on. Note that the directions of stimulation and suppression links in this network are opposite to those in the situation-oriented

immune network. This is because the effect-to-cause chain of behaviors is required for accomplishment of given demand. In other words, situation-oriented and goal-oriented immune networks play the roles of *data-driven* and *conceptually-driven* processing system, respectively.

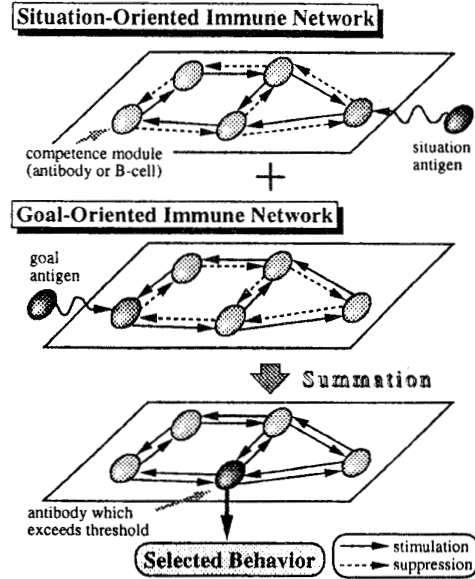


Fig.5: Immune networks for autonomous mobile robots.

3.3 Dynamics of antibody selection mechanisms

The total concentration of antibody i a_i is defined as follows:

$$a_i = (1 - \gamma_i) \cdot a_i^s + \gamma_i \cdot a_i^g \quad (1)$$

where a_i^s and a_i^g denote the concentration of i -th antibody in the situation-oriented and goal-oriented immune networks, respectively, and γ_i denotes the relative balance between data-driven and conceptually-driven processing. The concentration of i -th antibody in the situation-oriented immune network a_i^s is calculated as:

$$\frac{da_i^s}{dt} = \left\{ \frac{\sum_{j=1}^N m_{j,i} \cdot a_j^s}{N} - \frac{\sum_{k=1}^N m_{i,k} \cdot a_k^s}{N} + m_i - k_i \right\} \cdot a_i^s \quad (2)$$

where N denotes the number of antibodies in the concerned layer. The first term on the right hand side represents the activation level that other antibodies spread to antibody i (stimulation from other antibodies), the second term the activation level that antibody i spreads to other antibodies (suppression from other antibodies), the third term the external input of activation to antibody i (from antigens), and the forth term the dissipation (natural death). The concentration of i -th antibody in the goal-oriented immune network a_i^g can be calculated in the same way. $m_{j,i}$ and m_i denote matching ratios between antibody j and antibody i and between antibody i and antigen, respectively and are given by:

$$m_{ji} = \alpha \sum_k^L I_j(k) \oplus \overline{P_i(k)} \quad (3)$$

$$m_i = \beta \sum_k^L E(k) \oplus \overline{P_i(k)} \quad (4)$$

where α and β are positive constants, $\oplus$ represents *exclusive-or* operator, and L is the length of paratope, idiotope and epitope written in binary string forms. $I_j(k)$, $P_i(k)$ and $E(k)$ represent the k -th value of add and delete list (idiotope) of antibody j , that of precondition list (paratope) of antibody i , and that of external input list (epitope), respectively. Furthermore, to select the suitable antibody, we set two thresholds. If the concentration of the antibody exceeds the lower threshold, the antibody can spread its activation level to other antibody according to its matching rate. When the concentration of an antibody exceeds the higher threshold, the antibody is selected.

3.4 Comparison between our methods and Maes's approach

There are many correspondences between our proposed methods and Maes's approach. An agent in Maes's behavior network is described by a tuple (lp, la, ld, a) as shown in Fig.6. lp is a list of preconditions. la and ld are add and delete lists. In addition each agent has an activation level a . It can be seen that agents in Maes's behavior networks correspond to antibodies (or B-lymphocytes), and the precondition list of an agent can be regarded as a paratope of an antibody, the add list and the delete list as idiotope, and activation level as the concentration of antibodies, respectively (see Fig.6). Moreover, situation detected by agents and goals given to the society of agents works as antigen or epitope, and the interaction between agents corresponds to stimulation and suppression between antibodies.

However, the equations of antibody's selection based on the immunological dynamics are different from those of Maes's behavior networks. Namely, in our method quantity of spreading activation to the other antibodies are taken into account of both the concentrations of source and sink antibodies, whereas in Maes's network the above quantity considers merely the activation of the agent concerned. This difference of equation dynamics leads to the difference of behaviors discussed later.

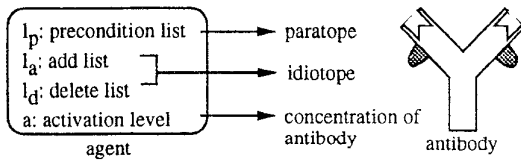


Fig.6: Relationship between Maes's agent and an antibody in immune system.

4. Application to behavior control of autonomous mobile robots

To confirm the validity of our proposed method in section 3, we apply this idea to obstacle avoidance

problem of autonomous mobile robot by carrying out simulations.

Fig.7 indicates the robot used in the following simulations. We equipped this robot with two kinds of detector: *obstacle detectors* and *goal detector*. Obstacle detectors are installed in eight directions shown in Fig. 7 and they can detect the existence of obstacles within heavy-shaded regions. In this study, detectable range of each sensor is assumed to be as three times as the radius of the robot. On the other hand, goal detector recognizes the direction of destination at any position.

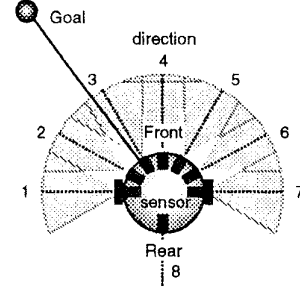


Fig.7: Autonomous mobile robot.

To be able to cope with the surrounding environment properly, we must set various kinds of antibodies (i.e., repertoires of simple behavior modules). As a rudimentary stage of investigation, in this study, we prepared the following eight kinds of simple behaviors as antibodies: *move forward*, *move backward*, *turn right* (30° , 60° , 90°), *turn left* (30° , 60° , 90°). Details of the prepared antibodies are listed in Fig.8. As mentioned in section 3, paratope corresponds to precondition which have to be fulfilled before the antibody can be selected, and idiotope to the expected situation of the antibody concerned implements its own action to real world. In addition, symbol #

definition of paratope, idiotope and epitope

goal direction (1 ~ 8 or #)

L 90° L 60° L 30° Front R 30° R 60° R 90° Rear

← obstacle exists : 1
otherwise : 0 or #

↑ approach goal: 1
otherwise : 0 or #

antibodies for obstacle avoidance

antibody 1	###Q#####	move forward
antibody 2	1111111111	move backward
antibody 3	#####Q###	turn right 30°
antibody 4	#####Q###	turn right 60°
antibody 5	#####Q###	turn right 90°
antibody 6	#####Q###	turn left 30°
antibody 7	#####Q###	turn left 60°
antibody 8	#####Q###	turn left 90°

#####Q###
Q111111111
#####Q###
#####Q###
1111111111
1111111111
#####Q###
#####Q###
#####Q###

antibodies for goal approach

antibody 9	#####Q###4#	move forward
antibody 10	#####Q###Q8#	move backward
antibody 11	#####Q###5#	turn right 30°
antibody 12	#####Q###6#	turn right 60°
antibody 13	#####Q###7#	turn right 90°
antibody 14	#####Q###3#	turn left 30°
antibody 15	#####Q###2#	turn left 60°
antibody 16	#####Q###1#	turn left 90°

#####Q###4#
#####Q###Q8#
#####Q###5#
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#####Q###7#
#####Q###3#
#####Q###2#
#####Q###1#

paratope

behavior

idiotope

Fig.8: Prepared antibodies.

denotes that the condition can be taken as either 0 or 1, that is, it can be regarded as not so important information. Therefore we determined $I_j(k) \oplus P_i(k) = E(k) \oplus P_i(k) = 0$ in equation (3) and (4) in the case of $P_i(k) = \#$ or $I_j(k) = \#$.

For the ease of understanding, we take the antibody 8 which plays a role of turning left 90° as an example. From the figure, it is understood that the behavior can become active when the robot detects the existence of obstacles from left 60° to right 60° and the non-existence of obstacles in left 90° as represented by paratope, and idiotope represents that the robot would detect the non-existence of obstacles in front and the existence of obstacles on its right hand side if the robot turns left 90° .

According to the above preparation, we carried out the following simulations. Fig.9 shows the simulation result in the case of fixed obstacle environment. In the same figure, the solid rectangles denote obstacles. From the figure, it is understood that the robot can cope with the given environments by selecting the suitable antibodies (simple behavior) at each situation (antigen).

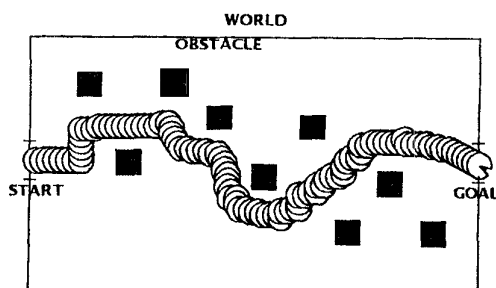
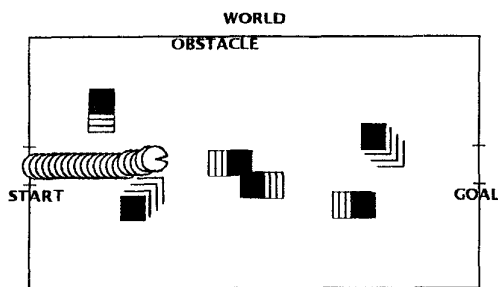


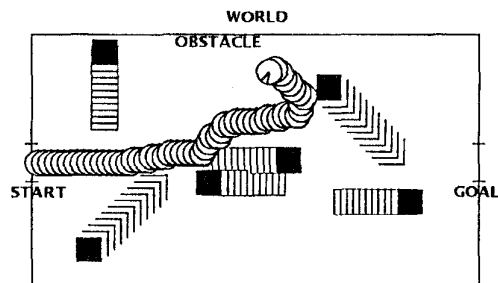
Fig.9: Simulation results in the case of fixed obstacles.

Fig.10 shows the simulation results in the case of moving obstacle environment. From the figures, it can be seen that the robot with immune networks can cope with these dynamically changing environments.

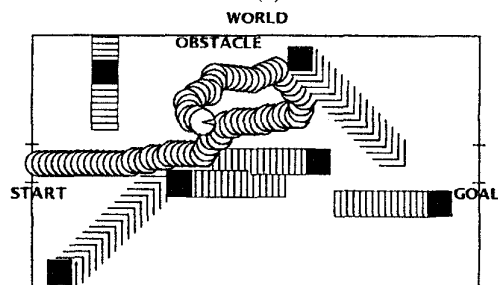
To demonstrate advantage of our method, we tried to compare our proposed method with Maes's behavior networks by carrying out simulations in same environments. Fig.11(a) illustrates simulation result using Maes's method. From this figure, the robot cannot reach the goal and fall into loops. On the other hands, Fig.11(b) shows result using immune networks. It is understood that the robot finds its way out of difficult impasse and reaches the goal successfully.



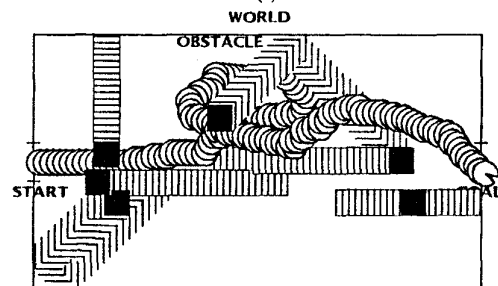
(a)



(b)



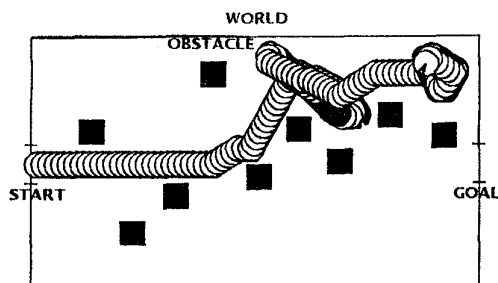
(c)



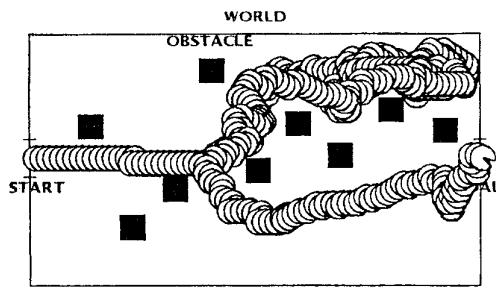
(d)

Fig.10: Simulation results in the case of moving obstacles ((a)->(b)->(c)->(d)).

To compare these two methods quantitatively, we tested these two methods by applying to 50 different randomly-arranged environments and recorded the number of successful reaching the goal and the number of action selections if succeeded. Fig.12 shows the relationship between average number of action selection and success ratio when we tried these tests 10 times. From the figure, it is comprehended that while our proposed method lead to more number of action selections, it keeps much higher success ratio than Maes's method.



(a) Maes's behavior network approach.



(b) Proposed immune network approach.

Fig.11: Simulation results of Maes's and our methods applied to the same environment.

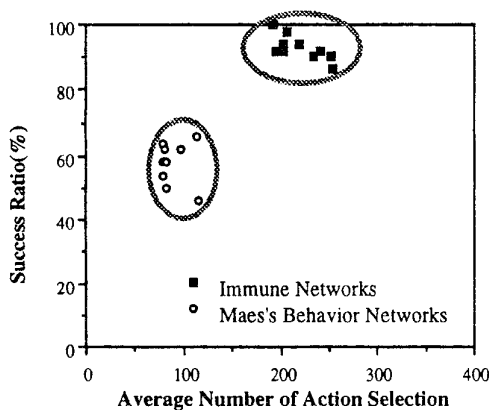


Fig.12: Quantitative comparison between immune and behavior network approaches.

5. Conclusions and Further Work

In this paper, we proposed a new behavior-based AI methodology from the immunological standpoint and confirmed the validity of our proposed system by applying to an obstacle avoidance problem of autonomous mobile robots. We obtained the result that our proposed method realized higher success ratio compared to Maes's behavior network approach. We are now trying to reveal this difference through quantitative and qualitative analyses.

Furthermore, it is worth mentioning that immune system has another important function called *meta-dynamics*. In living organisms, new species of antibodies are produced continuously from bone marrows. Our dream is to incorporate this meta-dynamics function into the current robot control method. Namely, if robots can not cope with the current situation, new antibodies are generated by meta-dynamic functions and obtaining immunity against new environment by incorporating them into the existing immune networks. Fig.13 schematically illustrates the concept of the meta-dynamics function. We are now considering this problem. Moreover, we will carry out experiments using real robots.

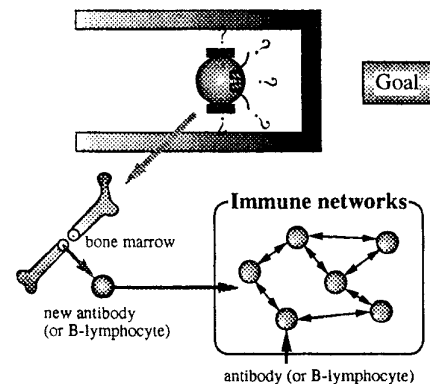


Fig.13: Meta-dynamics function.

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